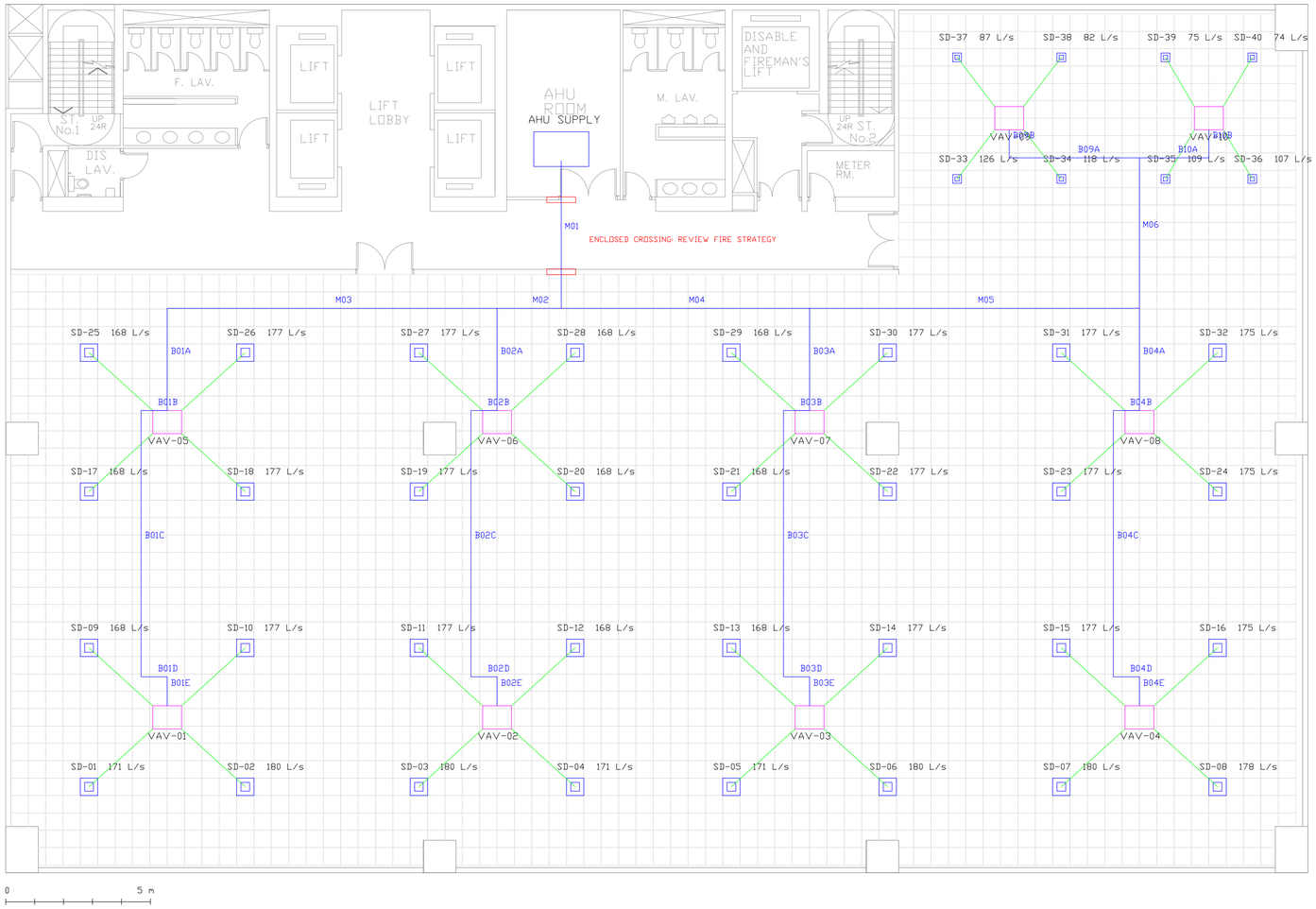


BSE3227 / HVACR 1

HVAC REVIEW LAYOUT v02

PRELIMINARY - NOT FOR SUBMISSION

40 SUPPLY DIFFUSERS / 10 VAV BOXES

4 outlets per VAV box

Printed office area: 1022 m2

Assumed total load: 120 W/m2

Assumed SHR: 0.75; room/supply 24/12 C

Peak supply: 6349.4 L/s

Minimum supply: 2539.8 L/s

BLUE: rigid supply duct / diffuser

GREEN: flexible connection

MAGENTA: VAV (schematic envelope)

All flex routes include 0.50 m allowance.

Maximum calculated flex: 3.474 m

Flex limit: 4 m (FSD 2023 cl. 7.2)

Assignment limit: 5 m

Duct sizes: width x height, mm

Main velocity <= 5.0 m/s

Branch velocity <= 4.5 m/s

Straight rigid friction <= 0.8 Pa/m

Rectangular aspect ratio <= 4:1

ADPI: CIRCULAR PATTERN TYPE 3

Generic louvered screen is RETIRED.

Peak-load ADPI remains unverified.

See independent classification audit.

Ceiling / beam clearances assumed;

coordinate before construction.

AHU corridor crossing requires fire

enclosure / boundary protection review.

Source architecture retained.

See CSVs for all 30 rigid sections.

RIGID SUPPLY DUCT SCHEDULE

ID	L/s	W x H (mm)	m/s	Pa/m
M01	6349.4	1500 x 850	4.98	0.207
M02	2771.5	800 x 700	4.95	0.320
M03	1385.8	700 x 400	4.95	0.509
M04	3577.9	1200 x 600	4.97	0.298
M05	2192.1	800 x 550	4.98	0.382
B01A	1385.8	700 x 450	4.40	0.374
B01B	695.9	450 x 350	4.42	0.563
B01C	695.9	450 x 350	4.42	0.563
B01D	695.9	450 x 350	4.42	0.563
B01E	695.9	450 x 350	4.42	0.563
B02A	1385.8	700 x 450	4.40	0.374
B02B	695.9	450 x 350	4.42	0.563
B02C	695.9	450 x 350	4.42	0.563
B02D	695.9	450 x 350	4.42	0.563
B02E	695.9	450 x 350	4.42	0.563
B03A	1385.8	700 x 450	4.40	0.374
B03B	695.9	450 x 350	4.42	0.563
B03C	695.9	450 x 350	4.42	0.563
B03D	695.9	450 x 350	4.42	0.563
B03E	695.9	450 x 350	4.42	0.563
B04A	1415.7	700 x 450	4.49	0.389
B04B	711.0	400 x 400	4.44	0.558
B04C	711.0	400 x 400	4.44	0.558
B04D	711.0	400 x 400	4.44	0.558
B04E	711.0	400 x 400	4.44	0.558
M06	776.4	450 x 350	4.93	0.689
B09A	412.6	400 x 250	4.13	0.670
B09B	412.6	400 x 250	4.13	0.670
B10A	363.8	350 x 250	4.16	0.726
B10B	363.8	350 x 250	4.16	0.726

Sizes apply to section IDs shown on plan.

Pressure gradient excludes fittings / flex.

TYPICAL FLOOR / SUPPLY AIR SINGLE-LINE PLAN

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